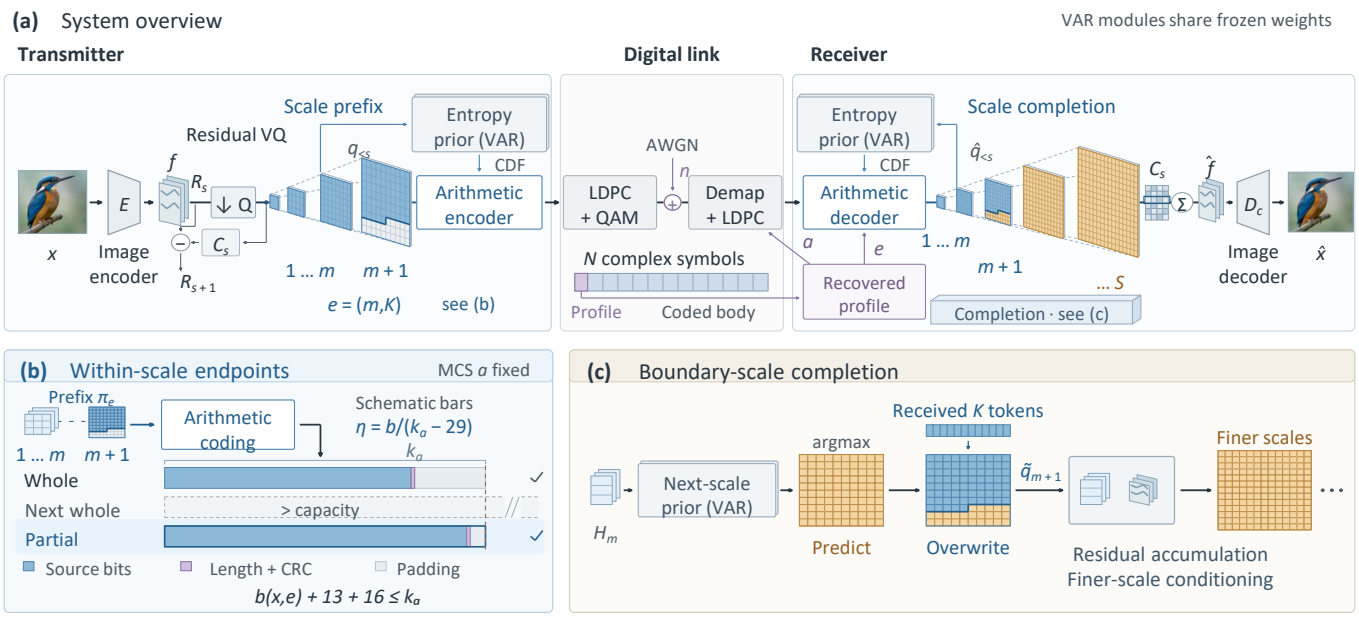

(a) System overview
VAR modules share frozen weights
Transmitter
Digital link
Receiver
Scale prefix
Residual VQ
Entropy prior (VAR)
CDF
Arithmetic encoder
Image encoder
x
E
f
R_s
↓ Q
C_s
R_{s+1}
q_{<s}
1 … m
m + 1
e = (m,K)
see (b)
AWGN
n
LDPC + QAM
Demap + LDPC
N complex symbols
Profile
Coded body
Entropy prior (VAR)
CDF
Arithmetic decoder
Scale completion
q̂_{<s}
a
e
1 … m
m + 1
… S
Recovered profile
Completion · see (c)
C_s
Σ
f̂
D_c
Image decoder
x̂
(b) Within-scale endpoints
MCS a fixed
Prefix π_e
1 … m
m + 1
Arithmetic coding
Schematic bars
η = b/(k_a − 29)
k_a
Whole
Next whole
> capacity
Partial
Source bits
Length + CRC
Padding
b(x,e) + 13 + 16 ≤ k_a
(c) Boundary-scale completion
Received K tokens
argmax
Finer scales
H_m
Next-scale prior (VAR)
Predict
Overwrite
q̃_{m+1}
Residual accumulation
Finer-scale conditioning
Tokens:
Transmitted / received
Predicted
Not transmitted